

7.

(1)

$$\mu_X(t) = 0 - 0 = 0$$

$$R_X(\tau) = E([B(t+1) - B(t)] [B(t+1+\tau) - B(t+\tau)])$$

$$= E[B(t+1, t+1+\tau) - B(t+1, t+\tau)]$$

$$= -R_B(t, t+1+\tau) + R_B(t, t+\tau)$$

$$\tau > 0 \text{ 时 } R_X(\tau) = t+1 - t - \min(t+1, t+\tau) + t$$

$$= t+1 - \min(t+1, t+\tau)$$

$$= \begin{cases} 1 - \tau & 0 < \tau < 1 \\ 0 & \tau \geq 1 \end{cases}$$

$$R_X(\tau) = \begin{cases} 1 - |\tau| & |\tau| \leq 1 \\ 0 & \text{其它} \end{cases}$$

(2)

$X(t)$ 是正态过程, 二阶矩过程,

$\mu_X(t)$ 常数。 R_X 只与 τ 有关

所以是宽平稳过程

(1)

$$\mu_X(t) = \sqrt{2} \cos t E(X) + \sin t E(Y)$$

$$E(X) = \int_{-1}^1 x(1-x) dx$$

$$= 0$$

$$\mu_X(t) = 0 \text{ 常数}$$

$$R_X(t, t+\tau) = 2 \cos t \cos(t+\tau) E(X^2) + \sin t \sin(t+\tau) E(Y^2)$$

$$E(X^2) = \int_{-1}^1 x^2(1-x) dx \quad E(Y^2) = \int_{-1}^1 \frac{1}{2} y^2 dy$$

$$= 2 \int_0^1 x^2(1-x) dx$$

$$= \frac{1}{6}$$

$$R_X(t, t+\tau) = \frac{1}{3} \cos(t+\tau - t) = \frac{1}{3} \cos \tau = R_X(\tau)$$

$$E(X^2) = R_X(0) = \frac{1}{3}$$

符合平稳过程

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$$(2) \quad \langle X(t) \rangle = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T}^T x(t) dt$$

$$= \lim_{T \rightarrow \infty} \frac{\sqrt{2} X}{T} \sin T$$

$$= 0$$

$$= \mu_X$$

均值具有各态历经性

$$(3) \quad \langle X(t)X(t+\tau) \rangle = \lim_{T \rightarrow \infty} \left(\frac{1}{T} \cos T + \frac{\cos T \sin 2T}{2T} \right) + \frac{XY \sin T \sin 2T}{\sqrt{2} T} + Y^2 \left(\frac{\cos T}{2} - \frac{\cos T \sin 2T}{4T} \right)$$

$$= X^2 \cos T + \frac{1}{2} Y^2 \cos T$$

$$\neq R_X(\tau)$$

不具有各态历经性

14.

$$(1) \quad \mu_X(t) = \frac{1}{2} \times 1 = \frac{1}{2}$$

$$\tau > 0 \text{ 时 } R_X(t, t+\tau) = E(A^2) [R_N(t+1, t+1+\tau) - R_N(t+1, t+\tau) + R_N(t, t+\tau)]$$

$$= \frac{1}{3} [t+2 - \min(t+1, t+\tau)]$$

$$= \begin{cases} \frac{2-\tau}{3} & 0 < \tau < 1 \\ \frac{1}{3} & \tau \geq 1 \end{cases}$$

$$R_X(\tau) = \begin{cases} \frac{2-|\tau|}{3} & -1 \leq \tau \leq 1 \\ \frac{1}{3} & \text{其它} \end{cases}$$

(2)

μ_X 常数, R_X 只与 τ 有关,

$E(X^2) = R_X(0) = \frac{1}{3}$ 是二阶矩过程

故宽平稳过程

$$(3) \quad \lim_{T \rightarrow \infty} R_X(\tau) = \frac{1}{3} \neq \mu_X^2 = \frac{1}{4}$$

不具有均值各态历经性